

The Thermodynamic Arrow

A Boundary-Conditioned Corollary of Realized History and Scope

1. Introduction

The thermodynamic Arrow presents a familiar mismatch. Macroscopic histories display pronounced temporal asymmetry, while many of the underlying physical descriptions admit substantially greater time symmetry. Boundary conditions, statistical typicality, entropy production, and cosmological structure address important parts of that mismatch. But once an asymmetric physical boundary has actually participated in a realized thermodynamic history, a narrower question remains: what makes later continuation remain answerable to that incorporated history without requiring the boundary to be physically re-imposed again and again?

The answer developed here is that no additional Arrow-maintaining mechanism is required for that job.

That claim is conditional. It does not explain why the relevant boundary is asymmetric, derive entropy increase, select a successor state, or assert one Arrow for every possible universe or every physical scope. Physics describes the boundary, thermodynamic asymmetry, realized history, and physical outcomes that actually obtain. The present argument concerns what follows structurally once those conditions are in place.

Once an asymmetric physical boundary has participated in a realized thermodynamic history, later continuation of that same history remains answerable to the incorporated passage through which it was realized. The history does not receive a free reset merely because later states lose microscopic information, become insensitive to earlier conditions, or admit coarser descriptions. This answerability does not prescribe what happens next. Multiple successors can remain available, and an accounted return to an earlier-equivalent state is not excluded.

At larger scales, no singular Arrow follows merely from the existence of local thermodynamic asymmetries. One Arrow at a declared scope follows only where the

physically described local orientations can be identified as the same relevant orientation and jointly supported across that domain. Where those conditions are satisfied, the thermodynamic Arrow at that scope is a conditional structural corollary of the realized architecture rather than an additional future-driving object. Where they fail, narrower Arrows may remain.

The result is also directly defeasible. Hold fixed the asymmetric boundary, realized macrohistory, thermodynamic asymmetry, lineage answerability, common orientation, and scope coherence. If a physically adequate case nevertheless requires a new irreducible Arrow-maintaining or future-driving object, process, or mechanism to obtain one Arrow at that same scope, the present corollary fails for that case.

The paper therefore separates questions often gathered under the single heading of the Arrow of Time. Why the relevant boundary is asymmetric, why irreversible behavior occurs or is statistically typical, why realized continuation remains answerable to incorporated history, and when local orientations compose as one Arrow are related questions, but they are not the same question. The argument that follows earns the narrower result stated above.

2. Boundary Conditions and the Arrow of Time

The thermodynamic Arrow is often introduced through a familiar contrast. Macroscopic processes display a pronounced temporal asymmetry: gases spread, temperature gradients dissipate, records accumulate in one direction, and isolated nonequilibrium systems ordinarily approach equilibrium rather than spontaneously departing from it. Yet many of the microscopic dynamical frameworks used to describe those systems permit time-reversed evolutions. The resulting problem is real, but it is not one problem. Questions about statistical typicality, special boundary conditions, cosmological structure, local entropy production, decoherence, and the relation between local and global temporal orientations can all appear under the same heading while asking for different explanatory objects.

Boltzmannian statistical mechanics addresses one central part of this landscape. Given a nonequilibrium macrostate and an appropriate statistical treatment of its compatible microstates, overwhelmingly many microscopic configurations can evolve toward macrostates occupying larger regions of phase space. Lebowitz emphasizes precisely

this statistical reconciliation between time-symmetric microscopic dynamics and time-asymmetric macroscopic behavior, while also emphasizing the importance of low-entropy conditions in accounting for the observed one-sided thermodynamic evolution.¹ The explanatory burden here concerns typical macroscopic behavior under physical dynamics and statistical assumptions. It does not, by itself, settle why the relevant boundary conditions have the asymmetry they do.

Past-Hypothesis approaches place that latter burden more explicitly at a cosmological boundary. In one influential formulation, a simple low-entropy condition on the early universe, combined with the dynamical laws and an appropriate statistical measure, is intended to account for a broad family of later temporal asymmetries. Albert continues to defend this general explanatory strategy, arguing from cosmological boundary conditions toward the temporal orientation of local systems.² But even within statistical mechanics and cosmology, the necessity of a conventional low-entropy initial-state hypothesis is not uncontested. Goldstein, Tumulka, and Zanghì examine cosmological models in which a thermodynamic Arrow can arise without simply imposing such a hypothesis at an initial time.³ The important point for the present paper is not to adjudicate that dispute. It is that the physical origin and exact form of the relevant asymmetry remain distinct from the question of what follows once a physical account describes an asymmetric boundary as obtaining.

Gravity makes that distinction harder rather than easier. In an ordinary gas, spatial uniformity is naturally associated with high entropy. In cosmology, a highly homogeneous early universe cannot be treated by simply carrying that intuition over unchanged, because gravitational degrees of freedom alter the entropy problem. Penrose's Weyl-curvature program is one influential attempt to characterize the specialness of the early gravitational state geometrically, and later work has continued

¹ Joel L. Lebowitz, "Macroscopic Laws, Microscopic Dynamics, Time's Arrow and Boltzmann's Entropy," *Physica A: Statistical Mechanics and its Applications* 194, nos. 1–4 (1993): 1–27, DOI: 10.1016/0378-4371(93)90336-3. Lebowitz develops the statistical reconciliation between microscopic reversible dynamics and macroscopic irreversible behavior and emphasizes the role of low-entropy initial conditions.

² David Z. Albert, "The Direction of Time: From the Cosmos to Local Systems," in Cristian López and Olimpia Lombardi, eds., *The Arrow of Time: From Local Systems to the Whole Universe* (Cambridge University Press, 2025), DOI: 10.1017/9781009581608.002. Albert presents a contemporary defense of the Past-Hypothesis strategy for connecting cosmological asymmetry to local temporal asymmetries.

³ Sheldon Goldstein, Roderich Tumulka, and Nino Zanghì, "Is the Hypothesis about a Low Entropy Initial State of the Universe Necessary for Explaining the Arrow of Time?" *Physical Review D* 94, 023520 (2016), DOI: 10.1103/PhysRevD.94.023520. The authors discuss cosmological models in which a thermodynamic Arrow can arise without imposing the conventional low-entropy initial-state hypothesis.

to investigate formulations of that proposal.⁴ There is no need here to choose a final measure of gravitational entropy. The relevant lesson is narrower: even the physical content of “a low-entropy beginning” is theory-sensitive once gravity is involved. A structural argument about the Arrow should therefore begin from the appropriate asymmetric boundary as a physically described condition, not smuggle a preferred cosmological entropy prescription into its premises.

Other Arrow programs begin still farther from a cosmological low-entropy postulate. Barbour, Koslowski, and Mercati exhibit a time-symmetric gravitational model whose typical solutions divide around a distinguished Janus point into two oppositely directed branches, challenging the idea that an Arrow always requires a specially imposed one-sided initial condition.⁵ Schulman constructed a model in which two weakly interacting systems maintain opposite thermodynamic arrows.⁶ Neither construction establishes that our universe realizes the corresponding architecture. They do establish something methodologically important: the existence of thermodynamic orientation in a system does not by itself license an unrestricted inference to one universal orientation over every physically or mathematically available region.

The same lesson appears from a different direction in nonequilibrium physics. Stochastic thermodynamics can characterize irreversibility at the level of trajectories by comparing forward and reverse path probabilities and relating those asymmetries to entropy production. Crooks’ fluctuation relation is a foundational example: for stochastic, microscopically reversible dynamics, the relation constrains the probability structure of forward and reverse nonequilibrium processes.⁷ More recent work has shown that evidence for a local Arrow can itself be decomposed among the irreversible dynamics of individual components and irreversibility arising from their

⁴ For one formulation of Penrose’s Weyl Curvature Hypothesis and its relation to the proposed low gravitational entropy of the early universe, see Ovidiu-Cristinel Stoica, “On the Weyl Curvature Hypothesis,” *Annals of Physics* 338 (2013): 186–194, DOI: 10.1016/j.aop.2013.08.002. The hypothesis is used here as an example of a live gravitational-boundary program, not as a settled definition of gravitational entropy.

⁵ Julian Barbour, Tim Koslowski, and Flavio Mercati, “Identification of a Gravitational Arrow of Time,” *Physical Review Letters* 113, 181101 (2014), DOI: 10.1103/PhysRevLett.113.181101. Their time-symmetric Newtonian model contains a distinguished Janus point separating two oppositely oriented branches.

⁶ L. S. Schulman, “Opposite Thermodynamic Arrows of Time,” *Physical Review Letters* 83 (1999): 5419–5422, DOI: 10.1103/PhysRevLett.83.5419. Schulman exhibits a model of weakly coupled systems maintaining opposite thermodynamic arrows.

⁷ Gavin E. Crooks, “Entropy Production Fluctuation Theorem and the Nonequilibrium Work Relation for Free Energy Differences,” *Physical Review E* 60 (1999): 2721–2726, DOI: 10.1103/PhysRevE.60.2721.

correlations.⁸ These are genuine physical measures of nonequilibrium asymmetry. They also make clear that “there is an Arrow” can mean something sharply local and operational, rather than automatically naming a property of the universe as an undifferentiated whole.

Quantum theory introduces still another separation. Decoherence supplies an effectively irreversible loss of accessible interference in subsystems through their interaction with environmental degrees of freedom, but the dynamical formalism does not by itself select the special low-correlation conditions from which the familiar decoherent orientation proceeds. Al-Khalili and Chen therefore formulate an Entanglement Past Hypothesis: a proposed low-entanglement boundary condition intended to play for the decoherent Arrow a role analogous to that played by the thermodynamic Past Hypothesis.⁹ Their analysis explicitly distinguishes the thermodynamic and decoherent arrows and leaves their unification as an open question. The distinction is useful here because it prevents “the Arrow of Time” from becoming a name for every temporal asymmetry at once.

Taken together, these approaches suggest a more discriminating starting point. The statistical question asks why entropy-increasing macroscopic evolution is typical or overwhelmingly probable under the relevant physical conditions. The cosmological question asks why the universe, or the physically relevant branch or region, has the asymmetric boundary structure required by a particular account. Nonequilibrium and quantum approaches ask how irreversibility is physically represented in trajectories, correlations, environments, and observables. Models with opposed or branch-relative arrows ask when apparently local orientations can legitimately be treated as one larger orientation. These questions interact, but answering one is not automatically an answer to the others.

The present argument therefore begins downstream of several of them. It does not derive a low-entropy universe, choose a gravitational entropy measure, replace Boltzmannian typicality, supply an entropy-production law, or explain decoherence. It

⁸ Christopher W. Lynn, Caroline M. Holmes, William Bialek, and David J. Schwab, “Decomposing the Local Arrow of Time in Interacting Systems,” *Physical Review Letters* 129, 118101 (2022), DOI: 10.1103/PhysRevLett.129.118101. The work decomposes local evidence of irreversibility into contributions from individual dynamics and correlations among components.

⁹ Jim Al-Khalili and Eddy Keming Chen, “The Decoherent Arrow of Time and the Entanglement Past Hypothesis,” *Foundations of Physics* 54, 49 (2024), DOI: 10.1007/s10701-024-00785-3. The authors distinguish the thermodynamic and decoherent arrows and propose a low-entanglement boundary condition for the latter.

begins from what the relevant physical account describes as obtaining at the scope under examination: a physically relevant asymmetric boundary condition, a descendant macrohistory from that boundary that has actually been realized, and the thermodynamic asymmetry attributed to that history. The remaining question is narrower. Once those conditions obtain, what makes later continuation of that realized history remain answerable to the boundary-conditioned passage it actually traversed?

That question is easy to obscure because discussions of the Arrow commonly move between possible histories, statistical ensembles, actual physical histories, records of those histories, and the boundary conditions under which they are modeled. The distinctions matter. A probability distribution over possible histories is not itself the history that occurred. A present macrostate is not identical to its full provenance. Loss of accessible information about earlier microstructure is not automatically loss of the fact that a particular physical history was realized. Before asking what follows from an asymmetric boundary, the object called a “realized history” therefore has to be made precise enough not to carry the conclusion by definition.

3. Realized History and the Residual Problem

A realized history is not introduced here as a new dynamical object. It is the actual physical history of the system under consideration as described by the relevant theory and evidence once a succession of physical states or processes has occurred. The adjective realized distinguishes that history from the wider set of dynamically admissible, statistically possible, counterfactual, or merely modeled histories. It does not imply that the realized history was the only history that could have occurred.

This distinction is elementary but consequential. Statistical mechanics routinely reasons over ensembles or measures on possible microscopic states while also describing the particular macroscopic evolution observed in a laboratory or attributed to the universe. A stochastic model assigns probabilities to alternative trajectories without erasing the distinction between the trajectory that occurs and those that do not. Quantum theory can retain several dynamically available alternatives even when a particular macroscopic record is the one physically under consideration. Nothing in the expression *realized history* therefore requires determinism. It identifies the history whose provenance is being claimed.

A second distinction concerns the physical substrate on which such a history is represented. An earlier Structural Flow analysis of time separated standing physical support from realized consequence and separated both from metric duration and thermodynamic orientation.¹⁰ A spacetime region can remain physically available without a new thermodynamic event being added merely because that region continues to exist. Conversely, a realized physical interaction can alter what is subsequently supported without becoming identical to spacetime itself. Likewise, a source/consequence relation can distinguish the roles of two configurations without thereby supplying an elapsed proper-time interval or a thermodynamic Arrow. The present paper inherits only those separations. It does not infer thermodynamic direction from a generic ordering relation, from spacetime structure alone, or from the mere fact that one event can be represented as consequentially downstream of another.

This matters because an asymmetric thermodynamic history contains more information than a bare ordering. Suppose a physical account describes a boundary condition (B) as thermodynamically asymmetric and identifies a realized descendant macrohistory (H) whose thermodynamic orientation is defined relative to that boundary. The physical theory must still do the substantive thermodynamic work: it must specify what counts as entropy, what dynamics apply, what statistical or quantum assumptions are legitimate, and how the relevant macroscopic asymmetry is represented and evidenced. Calling (B) a boundary and (H) its descendant does none of that work for free.

Once those physics-side burdens are satisfied, however, a different question appears. Later descriptions of the system need not retain all of the information contained in earlier descriptions. Coarse-graining deliberately suppresses microscopic detail. Equilibration can make many late observables insensitive to large classes of earlier microstates. Chaotic evolution can destroy practical reconstructibility. Decoherence can disperse phase information into inaccessible environmental correlations. An effective theory may no longer contain variables capable of representing much of the underlying earlier state. None of these facts is surprising, and none is challenged here.

¹⁰ *Time, Decomposed: Order, Consequence, History, and Measure Under Structural Flow, v1.2*, DOI: 10.5281/zenodo.22285454. This public antecedent distinguishes standing physical support, realized consequence, provenance/history, metric duration, and thermodynamic direction rather than treating them as one universal temporal structure. It specifically separates standing spacetime from realized physical history and source/consequence role-asymmetry from duration or a thermodynamic Arrow.

But predictive or observational insensitivity is not the same relation as historical provenance.

A gas near equilibrium may no longer permit reconstruction of the detailed molecular configuration from which it evolved. That does not by itself establish that the gas reached its present condition through no physical predecessor. A coarse-grained cosmological model may discard enormous quantities of early-universe microphysical information. That does not by itself turn the modeled late universe into a history with an unspecified or freely exchangeable origin. A record may be destroyed, inaccessible, or never formed at all while the physical process whose record is absent nevertheless occurred. What can be inferred from a present state and what physically belongs to the history of that state are different questions.

This distinction is especially important for the Arrow because the language of “memory” can otherwise do too much. A later state need not carry a recoverable microscopic imprint of every relevant feature of its boundary condition in order to belong to the history that descends from that boundary. Conversely, the existence of a macroscopic record is not what creates the historical relation. Records are physical structures within histories. They can provide evidence about provenance, and their own asymmetry can itself require explanation, but provenance is not reducible to the presence of a readable record.

The residual problem can now be stated without asking a boundary condition to act as a continuing force. If the relevant physical account describes an asymmetric boundary and a realized descendant history as obtaining, what would license later continuation of that same realized history to proceed as though the realized source relation and the accounted physical passage from it had never held? Put differently, does continued membership in a realized history require the original boundary to be dynamically re-imposed at each later stage, or is answerability to the incorporated passage already part of what makes that continuation the continuation of this history?

The question is not whether earlier physical conditions can influence later ones; that would be too weak to distinguish the present claim from ordinary dynamics. Nor is it whether every microscopic detail of an earlier state remains causally efficacious forever. It is a question about the validity conditions of a claim concerning one realized history. If a later macrostate is asserted to belong to that history, the assertion

inherits whatever accounted provenance is required to identify the history in the first place. If the provenance is discarded entirely, some additional account is needed of why the resulting object should still be regarded as the same realized history rather than a different history, an effective state description with unspecified ancestry, or a member of a broader equivalence class.

This is where the distinction between a boundary condition and a repeatedly acting boundary becomes decisive. Physical theories routinely specify boundary or initial data without introducing a second mechanism whose sole function is to remind every later state that the boundary once held. Dynamics determine what evolution is physically available; statistical structure determines what behavior is typical or probable; thermodynamic variables characterize the relevant macroscopic asymmetry. None of those facts alone answers the narrower provenance question, but neither do they obviously require a separate provenance-maintenance dynamics.

The problem is therefore not to find another process that pushes the thermodynamic Arrow forward. It is to determine whether such a process is required at all once the physical boundary, the realized descendant history, and the relevant thermodynamic orientation are described as obtaining. That is the point at which the present argument departs from the physics-side questions considered in the preceding section.

A final qualification is already necessary. A realized history is always claimed at some scope. A laboratory system, an open subsystem, a cosmological branch, and an entire cosmological model are not interchangeable domains merely because each exhibits an entropy gradient or another operational signature of irreversibility. Oppositely directed or independently oriented subsystems remain conceptually possible, and local measures of irreversibility do not automatically establish one thermodynamic orientation across a larger domain. The question of when local orientations can be treated as a single Arrow will therefore have to be earned separately. For the moment, the residual problem is narrower: given a physically described asymmetric boundary and the realized history it anchors at a declared scope, what follows from the fact that this history has that provenance?

That is the crossing on which the boundary-conditioned corollary turns.

4. The Thermodynamic Arrow as a Boundary-Conditioned Corollary

The residual problem is now narrow enough to state directly. Suppose the relevant physical account describes an asymmetric boundary condition, a realized descendant macrohistory, and the thermodynamic asymmetry attributed to that history as obtaining. The remaining question is not what dynamics generated the later states, what entropy measure applies, or why the boundary was asymmetric. It is what follows structurally once that physical passage has actually been realized.

The relevant constraint is not a new physical interaction. Structural Flow calls it Post-Authorization Answerability (PAA). In the form relevant to this paper, PAA means that once a transition has actually been realized and retained as part of a history, later lawful continuation begins from what that passage made true or from an accounted successor of it. A continuation cannot silently relocate its source to a condition prior to the realized passage while leaving the claim of historical continuity intact.¹¹

Applied to the thermodynamic Arrow, the point is modest but consequential. Let B denote the physically described asymmetric boundary under examination and let H denote the realized macrohistory descending from B. A later macrostate M claimed as part of H need not preserve a recoverable microscopic representation of B. It need not retain every boundary variable in an effective description. It need not remain predictively sensitive to the detailed microstate from which the history began. But if M remains part of H under a live PAA translation, its continuation remains structurally answerable to the accounted passage through which H was realized.

This is the sense in which the Arrow can be boundary-conditioned without treating the boundary as a continuously acting physical cause. The boundary is not required to recur, propagate itself as a separate field, or be dynamically re-imposed at each later macrostate. Physical dynamics still determine which successors are available. Statistical mechanics still carries the burden of typicality and probability. Thermodynamics still describes the relevant entropy and entropy-production behavior. PAA performs a different job: it keeps the incorporated realized passage live as a condition of lawful continuation without selecting the next physical outcome.

The corollary can therefore be stated as follows.

¹¹ *SF_Arrow_Corollary_Contract_and_Adapter_v0_3.lean*. The public Arrow contract encodes the lineage-facing PAA translation used here: lawful continuation remains structurally rooted in an accounted successor path from the realized macrohistory, while an unaccounted reset of the incorporated boundary passage is excluded.

Arrow Corollary

For a declared physical scope Σ , suppose the relevant physical account describes an asymmetric physical boundary B , a realized descendant macrohistory H rooted in B , and the thermodynamic asymmetry attributed to H as obtaining. If later continuation of H satisfies the PAA condition just stated, it remains structurally answerable to the accounted successor lineage and cannot receive an unaccounted reset of the incorporated passage. Where the physically described local thermodynamic orientations can also be identified as the same relevant orientation and jointly supported over Σ , one thermodynamic Arrow at Σ is a conditional structural corollary of the realized architecture rather than an additional future-driving or Arrow-maintaining object.

The scope clause is part of the result rather than a qualification added afterward. A realized asymmetric history in one laboratory system, subsystem, cosmological region, or branch does not by itself establish one orientation over a larger domain. The corollary therefore does not promote a local Arrow to a universal Arrow merely by changing scale. The larger claim enters only where the local orientations can be identified as the same relevant orientation and jointly supported over the declared scope.

Nor is the corollary an absolute irreversibility theorem. A realized history may contain fluctuations, losses, reversals of particular observables, or even a physically accounted return to a state equivalent in relevant respects to an earlier one. Such a return does not erase the passage by which it was reached. The distinction is between return and reset: a return through an accounted successor history remains part of that history, whereas an unaccounted reset treats an incorporated realized passage as though it had never occurred.¹²

The same constraint does not select a unique future. More than one successor can remain compatible with the same realized provenance. Which successor is physically realized remains a matter for the governing physical dynamics, probabilities, boundary

¹² Ibid. The formalization distinguishes an accounted return to a boundary-equivalent source from an unaccounted reset; it therefore does not encode absolute irreversibility.

conditions, interactions, and measurements. PAA governs answerability after realization; it does not choose the continuation.¹³

This separates two forms of inevitability that should not be conflated. The realized physical outcome was not prescribed. Once the passage has been realized and the PAA translation is live, however, answerability to that incorporated passage is not optional for later continuation of the same history. That is the structural inevitability asserted here. It does not follow that any particular successor macrostate, fluctuation, event, or microscopic trajectory had to occur. A history can be structurally source-bound while remaining dynamically open among multiple possible successors.

The result also leaves the physical origin of the asymmetric boundary open. That omission is substantive. The corollary begins only after the relevant physical account describes the boundary as obtaining in the form required by the case under analysis. A Past-Hypothesis formulation, a Janus-type model, a quantum-cosmological proposal, or a future physical framework may disagree about what the relevant boundary is, whether one exists in the required form, or how its thermodynamic significance should be characterized. The explanation of the boundary's origin therefore remains an independently open physical question.¹⁴

The resulting claim is therefore stronger than the observation that initial conditions influence later evolution, but narrower than a complete physical explanation of thermodynamic asymmetry. Once an asymmetric boundary has actually participated in a realized thermodynamic history, later continuation of that history remains structurally answerable to the incorporated passage. No second mechanism is required merely to make the boundary-conditioned history answerable to what it has already traversed.

That conclusion establishes the corollary. It does not yet establish how much thermodynamic content provenance itself can carry. In particular, it remains necessary to separate historical answerability from entropy production, microscopic memory, absolute irreversibility, and determinism.

¹³ Ibid. The formalization permits distinct successor branches under the same provenance constraint, preserving the distinction between source-bound continuation and unique-outcome selection.

¹⁴ Ibid. The formal representation leaves the physical origin of the asymmetric boundary open even where the conditional premises of the Arrow Corollary are otherwise admitted.

5. What the Corollary Does Not Supply

The reach of the corollary is clearest at the boundary between structural answerability and physical thermodynamics. The two should not be conflated. A realized history can remain answerable to an incorporated source relation without provenance generating the entropy gradient, determining a rate of entropy production, supplying a statistical measure, fixing a coarse-graining, or selecting the microscopic dynamics. Those remain physical questions. PAA governs the structural answerability of lawful continuation after realization; it does not convert that answerability into a thermodynamic mechanism.

This distinction matters because the physical source of thermodynamic asymmetry remains contested even after a boundary condition is specified. Contemporary work continues to locate thermodynamic orientation in different combinations of special conditions, open-system structure, dynamical assumptions, statistical descriptions, and equilibration behavior.¹⁵ Earlier criticism of the Past Hypothesis likewise emphasizes that even a well-defined low-entropy cosmological boundary would not by itself settle every question about why ordinary thermodynamic behavior is as robust as it is.¹⁶ The present corollary does not adjudicate those disputes. If the relevant physical account does not describe the required thermodynamic asymmetry as obtaining, realized provenance cannot repair the missing physical condition.

PAA therefore does not imply that entropy must increase along every successor passage. It does not prohibit a fluctuation toward lower entropy. It does not assign probabilities to such fluctuations. It does not establish a monotonic macroscopic variable, define equilibrium, or determine whether a particular system is open or closed. A history may remain fully answerable to its realized provenance while displaying any thermodynamic behavior permitted by the governing physics. The Arrow Corollary enters only where the relevant physical account describes the thermodynamic orientation required by the case as obtaining.

¹⁵ Bryan W. Roberts, “Does Thermodynamics Have a Reversibility Problem?” *Mind* (2026), DOI: 10.1093/mind/fzag008. Roberts argues that thermodynamic temporal asymmetry can arise through physically specific circumstances including non-conservative/open descriptions or special past conditions, rather than from thermodynamics generically imposing a universal Arrow.

¹⁶ John Earman, “The ‘Past Hypothesis’: Not Even False,” *Studies in History and Philosophy of Modern Physics* 37 (2006): 399–430, DOI: 10.1016/j.shpsb.2006.03.002. Earman challenges both the definition and explanatory sufficiency of a cosmological low-entropy Past Hypothesis, including the inference that such a boundary by itself explains the success of ordinary thermodynamics.

That separation also sharpens the objection that the corollary says nothing more than initial conditions matter. If the claim were merely that different initial conditions can lead to different later states, the objection would be decisive. Ordinary dynamics already describes that result. The present claim concerns something different: whether a history that has actually been realized through a particular boundary-conditioned passage can later continue as that same history after the incorporated source relation has been discarded without replacement.

The difference can be seen by separating the identity of a state from the identity of a history. A macrostate (M) may be reachable through more than one physically admissible history. Two histories can therefore terminate in states that are identical under the variables retained by a particular macroscopic description while differing in the physical passages by which those states were reached. Nothing in the corollary requires the state description itself to encode that difference. If the scientific claim concerns only the present macrostate (M), historical provenance may be irrelevant to the purpose of the model. If the claim is instead that (M) belongs to a particular realized history (H), provenance becomes part of the structural relation carried by that historical claim.

This prevents the argument from reducing to a definition. A realized history is not defined by requiring every later state to display its origin. Nor is a macrostate required to contain a hidden label identifying its ancestry. The same state-level description can be compatible with several histories precisely because state identity and historical identity need not coincide. The corollary enters only when one of those histories is the realized one under analysis. At that point, erasing the incorporated provenance while retaining the same continuation claim changes the structural status of the history.

Coarse-graining illustrates the difference. A coarse-grained description can legitimately map many microscopic histories onto the same effective state. It can erase distinctions that are irrelevant to the variables being modeled and can make later predictions insensitive to large portions of earlier microphysical detail. Nothing in the Arrow Corollary requires an effective model to recover information that its own construction has discarded. What does not follow from that compression is that every physically distinct history represented by the same coarse-grained state has thereby become one and the same realized history.

The same point applies to observational forgetting. The absence of a present record can limit what an observer is entitled to infer about a system's past. It can render competing histories empirically indistinguishable under the available evidence. In such a case, the proper conclusion may be that provenance is unknown or underdetermined. It is not that realized provenance has ceased to be a structural relation of whichever physical history actually occurred. Epistemic access to a history and structural answerability to its realized provenance are different questions.

This distinction also limits what can be inferred from source-binding. A history can be source-bound without being future-bound. Once a particular successor has been realized, continuation from that successor remains answerable to the passage by which it was reached. Before that successor is realized, however, PAA supplies no rule selecting it from the alternatives allowed by the physical theory. Stochastic branching, quantum alternatives, deterministic evolution, and other physical architectures can therefore differ radically in how successors are selected while remaining compatible with the same provenance constraint.¹⁷

The distinction is important enough to state in another form. Structural inevitability is retrospective and conditional; it is not prospective outcome necessity. Nothing in the corollary says which admissible passage had to be realized beforehand. Once a passage has been realized and retained under the live PAA translation, however, later continuation of that same history cannot erase it for free. The inevitability concerns answerability after realization, not selection before realization.

Nor does provenance itself supply temporal direction. A source relation can distinguish how a realized history was constituted, but thermodynamic orientation enters this paper only where the relevant physical account describes a thermodynamic asymmetry as obtaining. Without that physical asymmetry, PAA can still govern realized provenance, but no thermodynamic Arrow follows through the route developed here. The direction relevant to the Arrow is therefore not manufactured by structural answerability.

The division of labor is correspondingly precise. Physics describes the boundary, realized processes, thermodynamic variables, dynamics, physical asymmetry, and particular outcomes that actually obtain. Structural Flow supplies the structural

¹⁷ *SF_Arrow_Corollary_Contract_and_Adapter_v0_3.lean*. The public contract permits distinct successors under one admitted lineage, so provenance answerability does not encode unique-future selection.

grammar under translation. Where PAA is physically translated and live, it explains why a realized outcome remains answerable to the history it actually traversed without selecting that outcome or replacing the physical mechanisms that produced it.

With that limit in place, the remaining problem changes scale. Even if one realized system displays a boundary-conditioned thermodynamic orientation, it does not yet follow that another system, region, or branch shares that orientation. The next question is therefore not how provenance produces an Arrow, but under what conditions distinct physically described local orientations can be identified and jointly supported as one Arrow at a larger scope.

6. From Local Asymmetry to an Arrow at Scope

A local thermodynamic asymmetry does not become a larger-scale Arrow merely by being observed in more than one place. This is true even when each local result is individually well warranted. A laboratory system can display measurable entropy production; a subsystem can exhibit trajectory-level irreversibility; a cosmological branch can possess an orientation away from a distinguished boundary; and another region can display its own thermodynamic asymmetry. None of those facts, taken separately, establishes that all of the corresponding orientations are instances of one and the same Arrow.

The distinction is already visible in existing physics. Lynn and colleagues show that evidence for a local Arrow can be decomposed within interacting systems into irreversibility associated with individual variables and additional contributions arising from their correlations.¹⁸ Schulman's opposite-arrow construction goes farther in the other direction, exhibiting weakly coupled systems with opposed thermodynamic orientations.¹⁹ Janus-type models similarly make branch-relative orientation conceptually explicit.²⁰ These results arise from very different physical programs, but they share one methodological consequence: the scale at which an asymmetry is established and the scale at which one common orientation is claimed are not automatically identical.

¹⁸ Christopher W. Lynn, Caroline M. Holmes, William Bialek, and David J. Schwab, "Decomposing the Local Arrow of Time in Interacting Systems," *Physical Review Letters* 129, 118101 (2022), DOI: 10.1103/PhysRevLett.129.118101.

¹⁹ L. S. Schulman, "Opposite Thermodynamic Arrows of Time," *Physical Review Letters* 83 (1999): 5419–5422, DOI: 10.1103/PhysRevLett.83.5419.

²⁰ Julian Barbour, Tim Koslowski, and Flavio Mercati, "Identification of a Gravitational Arrow of Time," *Physical Review Letters* 113, 181101 (2014), DOI: 10.1103/PhysRevLett.113.181101.

That consequence cuts both ways. Locality alone does not show that orientations are unrelated. A common cosmological structure, shared boundary condition, interaction network, or other physically described relation may provide good reason to investigate whether distinct local asymmetries participate in one larger orientation. Indeed, global-to-local programs in the Arrow literature have explicitly investigated conditions under which a cosmological time-asymmetry can ground a common orientation in local irreversible processes.²¹ The point here is not that such a move is illegitimate. It is that the move has conditions.

The first condition is one of identity across variation. Two local systems need not realize irreversibility through identical mechanisms, variables, entropy measures, or effective descriptions in order to share an orientation. But if their local asymmetries are to count as instances of one Arrow, the physical description must provide a basis for identifying the direction in one case as the same relevant direction in the other. Mere use of the words past and future in both descriptions does not establish that identity, because those labels may already presuppose the orientation at issue.

The burden becomes especially clear when the systems differ in scale or description. A stochastic trajectory-level Arrow, an ordinary thermodynamic entropy gradient, a decoherent orientation, and a cosmological asymmetry may be related, but they are not made identical simply by pointing in what an observer conventionally calls the same temporal direction. Their common orientation must be grounded in the physical description rather than supplied by terminology alone. In some domains that relation may be direct. In others it may be model-dependent, approximate, or absent.

The paper therefore treats sameness of orientation across relevant variation as a condition that must be supported by the physical description, not inferred from shared vocabulary, common temporal labels, or the mere coexistence of local asymmetries.

Even successful identification is not yet enough. A collection of local orientations can be pairwise interpretable as pointing the same way while still failing to support one

²¹ Olimpia Lombardi, “Following Earman’s Time Direction Heresy: From the Global Arrow of Time to Local Irreversible Processes,” in Cristian López and Olimpia Lombardi, eds., *The Arrow of Time: From Local Systems to the Whole Universe* (Cambridge University Press, 2025), pp. 225–249, DOI: 10.1017/9781009581608.012. Lombardi reviews a global, non-entropic program relating cosmological time-asymmetry to local irreversible processes and examines the extent to which an Arrow can be defined when global conditions such as time-orientability, cosmic time, or time-asymmetry are unavailable.

Arrow over a proposed larger domain. The second burden concerns whether those locally identified orientations can be held together consistently across the whole declared scope. A region in which the orientation changes, bifurcates, becomes undefined, or conflicts with another physically supported orientation can break the larger claim without erasing the valid local results.

The larger claim therefore requires more than local comparability. The relevant local orientations must also be jointly supportable as one orientation over the entire declared scope. The scope itself is therefore part of the proposition. A statement that “the thermodynamic Arrow exists” is incomplete unless the domain over which the singular Arrow is asserted is either explicit or already fixed by context.

This yields an important asymmetry in the logic of scale. Failure at a larger scope does not propagate downward automatically. If one proposed cosmological domain contains regions whose thermodynamic orientations cannot be warranted as one common orientation, the appropriate conclusion is not that no thermodynamic Arrow exists anywhere. It is that the singular Arrow has not been established at that larger scope. Individual regions, subsystems, or coherent partitions may still possess well-defined Arrows.

Conversely, success at a smaller scope does not propagate upward automatically. Ten laboratories exhibiting the same direction of entropy production do not by themselves establish a universal thermodynamic orientation across every cosmological region or every mathematically allowed branch. The broader claim requires evidence or theory connecting those local results across the additional domain being claimed.

Scope therefore functions as a substantive part of the Arrow claim rather than as a hedge on it. Consider two candidate scopes, one nested inside the other. The smaller scope may support a common orientation because every relevant local asymmetry within it is comparably identified and jointly compatible. Enlarging the domain can introduce a new region whose orientation is unestablished or opposed. Nothing about the success of the smaller claim licenses silently importing that success into the larger one. The Arrow can therefore be stable under one scope and unavailable at another without contradiction.

This matters particularly for cosmological language. Expressions such as *the Arrow of Time of the universe* can suggest a universality stronger than the physical argument has

earned. A cosmological model may warrant such a global statement; another may warrant branch-relative arrows; another may leave the global relation open. Lombardi's recent treatment of global and local Arrow questions explicitly examines whether an Arrow remains definable when conditions such as global time-orientability, cosmic time, or global time-asymmetry fail, illustrating how strongly Arrow claims can depend on the structure and extent of the domain under consideration.²²

The present corollary therefore treats scope as something that must be established rather than as a conclusion obtained for free from local asymmetry. Where the physically described local thermodynamic orientations can be identified as the same relevant orientation across their variation, and where those orientations can be jointly supported over the declared domain, one thermodynamic Arrow at that scope becomes a conditional structural corollary of the realized architecture. Where either step fails, the larger singular Arrow is not established.

This is not a prediction that our universe contains a fractured thermodynamic orientation. It is a restriction on the inference: success at one scope does not by itself license an unrestricted universal Arrow.

The practical consequence is simple but far-reaching. There need not be one answer to the question "Does the Arrow exist?" independent of domain. The scientifically sharper question is: At what scope can the physically described thermodynamic orientations be supported as one common orientation? A local experiment, an interacting subsystem, a cosmological branch, and the full physically presented universe may receive different answers without any of those answers being incoherent merely because their scopes differ.

The title claim of this paper should therefore be read literally. The thermodynamic Arrow is a boundary-conditioned corollary of realized history and scope. Realized provenance carries boundary-conditioned answerability; scope determines how far the physically described orientations can be supported as one common Arrow. Neither burden licenses the other for free.

With the positive result and its scope now specified, the remaining task is to make clear how the corollary can fail. Those failure conditions are not peripheral caveats.

²² Ibid.

They identify the physical and structural observations that would block entry into the result or reopen a previously warranted claim.

7. Failure and Reopening

A conditional corollary is scientifically useful only if the conditions under which it stops applying are as clear as the conditions under which it enters. For the present result, three broad crossings can block or narrow the Arrow: the required physical conditions may fail to obtain, realized-lineage answerability may fail, or the local orientations may fail to support one Arrow at the proposed scope. The positive claim also carries one direct additional falsifier: even with those conditions retained, a genuinely new irreducible Arrow-maintaining or future-driving job may prove necessary.

The first crossing concerns the physical conditions described for the case. The corollary requires an asymmetric physical boundary, a realized descendant macrohistory, and a thermodynamic asymmetry associated with that history. If the relevant physical account does not describe the required boundary or thermodynamic asymmetry as obtaining, the corollary does not enter through that route. Structural answerability cannot manufacture a missing thermodynamic condition.

This distinction separates failure of entry from refutation of the conditional structural inference. Uncertainty about the early universe, disagreement about gravitational entropy, or an unresolved choice between competing cosmological boundary models does not by itself falsify the corollary. It means that one or more physical conditions remain unsettled. If the physical description changes, the domain verdict can change with it.

The second crossing concerns realized history. The corollary applies only where the states being related actually belong to the realized descendant history under analysis. If later evidence shows that two states previously treated as members of one macrohistory belong instead to different branches, different physical histories, or histories whose relation was only assumed, then the lineage translation connecting them no longer applies. Nothing stronger follows. Each history may still exhibit its own thermodynamic orientation.

There is, however, a sharper lineage failure. The lineage-answerability claim would be directly defeated if a physically adequate case retained the assertion that a later state belongs to the same realized history while the incorporated boundary passage and every accounted successor route connecting the later state to that passage could be removed without replacement. Such a case would show that the proposed source-answerability condition is unnecessary to lawful continuation of that realized history.

That is a stronger challenge than loss of memory, coarse-graining, practical irreconstructibility, or observational underdetermination. Those phenomena can prevent recovery of the past without establishing that the realized history is structurally free of the passage it actually traversed. The direct challenge has to preserve the continuation claim while eliminating the structural answerability relation.

The distinction matters because failure should not be manufactured by changing the object under discussion. If a later effective model intentionally represents only a present macrostate and makes no claim about which realized history produced it, provenance may properly disappear from that model. That does not contradict the corollary because the historical continuation claim has been dropped with it.

The third crossing is scope. Even where individual systems possess well-supported thermodynamic orientations, those orientations may fail to support one Arrow over a larger domain. They may fail to be identifiable as the same relevant orientation across their differences, or they may fail to remain jointly supportable over the proposed scope. Opposite-arrow constructions, branch-relative models, and cosmologies lacking the required global relations therefore pressure the scope of the claim without erasing narrower local results.

A scope failure is not a failure of thermodynamics in the constituent systems. Nor does it erase local Arrows that remain available at narrower scopes. It changes only how far one common orientation can be carried. A cosmological region may support one Arrow even if a larger model does not; two branches may each support an Arrow while failing to support one common Arrow across both.

The positive claim also has an explicit prove-me-wrong condition. Hold fixed the physically described asymmetric boundary, realized macrohistory, thermodynamic asymmetry, the requirement that later continuation remain answerable to the

incorporated realized passage without a free reset, the identification of the relevant local orientations as the same orientation, and their joint support across the same declared scope. If one thermodynamic Arrow nevertheless fails unless a genuinely new irreducible physical object, process, or mechanism performs an Arrow-maintaining or future-driving job not already carried by the physical description or by those relations, then the no-extra-driver claim fails for that case. Merely renaming ordinary dynamics, entropy production, decoherence, stochastic evolution, or other already-described physics as an “Arrow driver” does not satisfy this falsifier.

This makes the scope and applicability of the corollary empirically and theoretically revisable. New observations may reveal previously unknown regions or interactions. A cosmological model may change its account of global structure. A better treatment of a local irreversible process may change whether two orientations translate as the same. Conversely, new evidence may establish a relation that was previously missing and thereby support a broader scope.

Reopening is therefore part of the intended posture of the result. A new microscopic mechanism for entropy production can enrich the physical description without altering the lineage-answerability result. A new cosmological model can change the boundary condition without changing the conditional structural inference. Evidence for genuinely opposed local orientations can narrow the Arrow without defeating narrower ones. A same-history free reset would pressure the lineage-answerability claim directly. A qualifying additional-driver counterexample would defeat the no-extra-driver claim directly.

The result is therefore conditional but exact. Where the required physical conditions obtain, later continuation remains answerable to the incorporated realized passage without a free reset, and the relevant local orientations can be identified as the same orientation and jointly supported across the declared scope, one thermodynamic Arrow at that scope is a structural corollary of the realized architecture. The realized outcome was not prescribed. Its structural answerability after realization is not optional under those conditions. If one of the required conditions fails, the result does not enter or contracts; if a qualifying additional-driver counterexample is established, the present corollary fails for that case.

That failure structure is also what makes the result suitable for formal representation. The next section asks whether the successful route, open physical questions, declared failure routes, surviving narrower scopes, multiple possible successors, and explicit additional-driver falsifier can all be represented without changing the conditional character of the claim.

8. Formal Representation and Conditional Closure

The argument developed above is conditional in several distinct senses. A required physical condition can remain unresolved or fail to obtain. The realized-lineage translation can fail while other physical structure remains intact. A larger scope can fail while narrower Arrows survive. More than one future can remain compatible with the same realized provenance. And the no-extra-driver claim can itself be defeated by a qualifying same-scope counterexample. A formal representation is useful here only if it preserves those distinctions rather than converting them into a single binary claim.

For that reason, the Arrow Corollary was encoded as a machine-readable conditional contract in Lean and checked against the existing Structural Flow Universal Kernel.²³ The formalization does not derive the physical boundary, entropy, thermodynamic asymmetry, realized history, or cosmological scope. Those remain matters described, modeled, measured, or revised by the relevant physical account. The machine layer asks what follows structurally when the declared domain conditions and translations are instantiated.

The contract represents a successful Arrow translation only where the required physical conditions and Structural Flow translation conditions close together. It also represents cases in which a required physical condition remains open. An unresolved boundary condition is therefore not treated as a negative physical result; it leaves the corresponding Arrow claim unresolved. By contrast, violation of a required condition blocks the conditional inference through that route.

This distinction between open and failed matters scientifically. An incomplete cosmology should not be formalized as though it had established the absence of an

²³ *SF_Arrow_Corollary_Contract_and_Adapter_v0_3.lean* and *Structural_Flow_Universal_Kernel_v1.0.lean*. The Arrow-specific contract encodes the conditional domain and structural translation against the reusable Universal Kernel. Version 0.3 additionally gives `noExtraArrowDriverRequired` an explicit same-scope failure world while holding the other admitted conditions fixed.

asymmetric boundary. Nor should a known failure be relabeled as mere uncertainty. The machine representation preserves that difference so that changes in physical knowledge can alter the domain verdict without changing the structural grammar of the corollary.

The same separation applies to realized provenance. The formal representation blocks the Arrow route when the required descendant-history relation is violated and separately blocks continuation that requires an unaccounted reset of the incorporated passage. It does not infer from either condition that thermodynamics itself has failed. Physics describes the thermodynamic content and realized outcomes; the structural translation represents the answerability relations that become live through that realized architecture.

Scope is represented with the same discipline. If the conditions required for one common orientation fail at a proposed larger domain, the formal result does not infer that all local thermodynamic asymmetries disappear. It blocks only the translation to one singular Arrow at that scope. Narrower domains can remain independently viable.²⁴ This preserves the paper's central scope claim in executable form: success at one scale neither automatically propagates upward nor disappears merely because a broader claim fails.

The formalization also preserves the distinction between source-bound history and unique-future selection. A finite witness permits distinct successor branches under the same PAA-compatible provenance condition.²⁵ Machine closure therefore does not encode determinism. It checks that structural answerability can remain live while multiple successor outcomes remain available.

Version 0.3 of the Arrow adapter also gives the no-extra-driver claim its own explicit failure world. The other admitted conditions are held fixed while `noExtraArrowDriverRequired` alone is violated.²⁶ If a physically adequate same-scope counterexample requires a genuinely new irreducible Arrow-maintaining or future-driving job, the formal translation is blocked for that case. The contract does

²⁴ *SF_Arrow_Corollary_Contract_and_Adapter_v0_3.lean*. The formal scope-fracture case blocks one singular Arrow at the proposed larger scope while preserving the possibility that narrower partitions remain independently viable.

²⁵ Ibid. The formalization includes a finite case in which distinct successor branches remain compatible with the same admitted provenance condition, so conditional closure does not imply unique-future selection.

²⁶ Ibid.

not assert that such a counterexample exists, and ordinary already-described dynamics do not satisfy the seat merely by being relabeled as an Arrow driver.

The resulting machine verdict should be interpreted narrowly. The v0.3 Arrow adapter closes against the unchanged compatible Structural Flow Universal Kernel, with no kernel modification required.²⁷ This establishes that the encoded conditional structure, including its successful, open, and declared failure routes, is internally representable against that kernel.

It does not establish that the physical conditions are true of the universe. Lean has not measured an early-universe entropy condition, established a cosmological boundary, observed a realized macrohistory, determined an entropy-production law, selected a successor outcome, or shown that one thermodynamic orientation spans the largest physically relevant scope. Those remain physical and empirical questions.

The relationship between the scientific paper and the formal package is therefore one of constraint rather than substitution. The paper states the physical conditions, structural distinction, conditional corollary, and falsifiers in scientific language. The contract encodes that structure in a form that can be checked for consistency and for preservation of its stated limits. A mismatch between the two would be informative: it would indicate either that the prose has broadened beyond the encoded result or that the formal representation has failed to capture a distinction the scientific argument requires.

Machine-readable conditional closure is useful precisely because it remains reopenable. New physics can change which physical conditions are described as obtaining, whether the domain supports the required translation, or what scope closes as one Arrow. Such changes can alter the contract verdict without requiring the underlying formal kernel to change. A kernel revision would become relevant only if a new case exposed a defect in the reusable structural machinery itself rather than a changed domain instance.

The formal layer therefore adds no new Arrow-of-Time mechanism. Its function is to make the conditional boundaries of the claim executable and inspectable: rooted

²⁷ Ibid.; *Structural_Flow_Universal_Kernel_v1.0.lean*. The v0.3 Arrow adapter closes against the unchanged compatible Universal Kernel with no kernel modification required. The appropriate interpretation is conditional machine closure of the encoded successful, open, and failure routes, not empirical confirmation of the physical conditions.

continuation, no free reset, lawful return, multiple available successors, scope fracture, and the explicit additional-driver falsifier. The scientific result remains the one developed in the preceding sections.

9. Conclusion

The thermodynamic Arrow becomes more tractable when the questions gathered under that name are not required to share one explanation. Why a physically relevant boundary is asymmetric, why entropy-increasing behavior occurs or is typical, how irreversible processes are represented locally, and how far one common thermodynamic orientation extends are related problems, but they are not the same problem. Once those questions are separated, a narrower structural problem remains: what follows after an asymmetric physical boundary has actually participated in a realized thermodynamic history?

The answer developed here is a conditional structural corollary rather than another mechanism of irreversibility. Where the relevant physical account describes the asymmetric boundary, realized descendant history, and thermodynamic asymmetry as obtaining, later continuation of that same realized history remains structurally answerable to the incorporated passage. The history does not receive a free reset merely because later states no longer preserve a recoverable representation of the boundary.

That answerability is structurally inevitable in a strictly non-guarantee sense. Nothing in the structural claim selects the asymmetric boundary, the realized successor, the entropy history, or any later physical outcome. Those outcomes are the outcomes that physics describes as realized. Once a passage has been realized, however, continuation of that same history cannot proceed as though the incorporated passage were absent while retaining the same historical claim. The inevitability concerns answerability after realization, not outcome selection before it.

Nothing in that result transfers the work of thermodynamics to provenance. The corollary does not explain why the boundary is asymmetric, define entropy, produce entropy increase, assign fluctuation probabilities, determine microscopic dynamics, or select among competing cosmological accounts. Those remain physical questions.

The result also leaves the future open. More than one successor can remain physically available under the same realized provenance. A fluctuation may occur. An accounted return may occur. Different physical theories may describe successor selection differently. Historical answerability does not choose among them; it excludes only the free erasure of an incorporated realized passage while the same continuation claim is retained.

Scope supplies the second part of the corollary. Local thermodynamic orientations do not automatically constitute one universal Arrow. Where the physically described local orientations can be identified as the same relevant orientation and jointly supported over a declared scope, one thermodynamic Arrow at that scope is a conditional structural corollary of the realized architecture. Where those conditions fail, narrower Arrows may remain while the larger singular Arrow does not.

The resulting claim is restricted to this universe as presented and to scopes that satisfy the required conditions. It says neither that every possible universe has an Arrow nor that one Arrow extends across every branch, region, or mathematical solution. A different physical architecture can change whether the corollary applies, fails, or survives only at a narrower scope.

The no-extra-driver claim is explicitly defeasible. If a physically adequate same-scope case retains the asymmetric boundary, realized macrohistory, thermodynamic asymmetry, no-free-reset answerability, identification of the relevant local orientations as the same orientation, and their joint support across that scope, yet one Arrow still fails unless a genuinely new irreducible Arrow-maintaining or future-driving job is added, the present corollary fails for that case. The result therefore does not protect itself by definition.

The accompanying Lean contract does not establish the physical facts of the universe. It represents the conditional success and failure structure—including no-free-reset answerability, lawful return, multiple successor possibilities, scope fracture, and the explicit additional-driver falsifier—in a machine-checkable form. Whether the required physical conditions obtain remains a question for physics.

The thermodynamic Arrow reached here is therefore neither a prescribed future nor an additional physical object. The physical outcomes were not guaranteed. But once the relevant history is realized, its structural answerability to what it actually traversed

is not optional under the stated conditions. Where the physically described orientations can also be supported as one over a declared scope, the singular Arrow at that scope follows as the conditional structural corollary developed here.

AI Use Disclosure

Artificial intelligence tools were used during the development of this work for research assistance, adversarial review, conceptual organization, drafting support, editing, and formalization support. The author directed the research program, selected and evaluated the scientific claims and sources, determined the argument and its scope, and takes responsibility for the final content.

The Thermodynamic Arrow

A Boundary-Conditioned Corollary of Realized History and Scope

Signed,

Rick